Nuclear Physics

MAIS Physics & Astronomy Club (PAC) · Olympiad lecture + problem handout

8 lecture sections · 15 problems · 1 interactive experiment · 2 vector figures

Start with a physical picture, then introduce only the mathematics needed to make it predictive. The lecture connects everyday reasoning to astronomical measurements. Two vector diagrams and an interactive experiment support the argument; 15 graded problems and independent worked solutions turn the ideas into practice.

▼ Contents & learning path

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2. Missing mass is binding energy
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1. The nucleus in symbols

A nuclide ${}^{A}_{Z}X$ has Z protons and $N = A - Z$ neutrons. Chemical identity is set by Z; isotopes share Z but differ in N. Nuclear reactions conserve charge and nucleon number (and, more fundamentally, energy, momentum and other quantum numbers).

Alpha decay lowers A by 4 and Z by 2. Beta-minus decay converts a neutron into a proton, emitting an electron and an antineutrino, so A stays fixed and Z rises by 1. Gamma emission changes nuclear energy state without changing A or Z.

Use $1u = 931.494\ \mathrm{MeV}/c^2$, $N_A = 6.02214076 \times 10^{23}\ \mathrm{mol}^{-1}$, $c = 2.99792458 \times 10^{8}$ m/s and one year $= 3.15576 \times 10^{7}$ s.

2. Missing mass is binding energy

A bound nucleus weighs less than its separated constituents. The mass defect is

$$\Delta m = Z m_p + N m_n - m_{\text{nucleus}}, \qquad B = \Delta m c^2. \tag{1}$$

When using neutral atomic masses, replace proton masses with hydrogen-atom masses for reactions whose total electron count balances; electron masses then cancel to excellent accuracy. Binding energy per nucleon B/A measures how tightly bound a nucleus is.

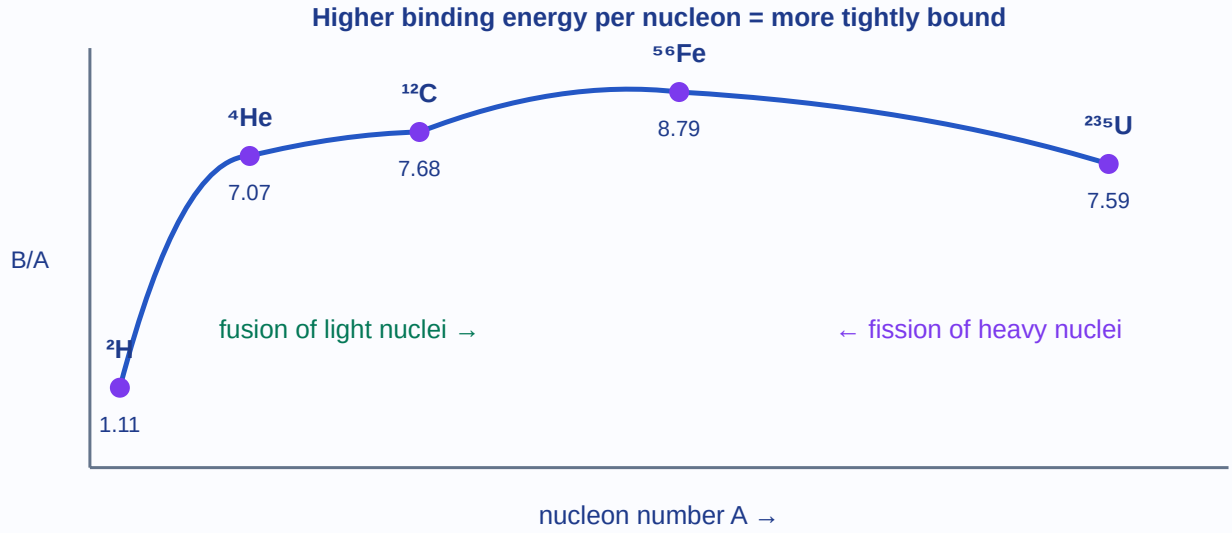


Figure 9. Selected approximate values in MeV per nucleon; the connecting curve is schematic. Energy release is determined by the actual initial and final masses, not by the curve alone.

3. Why fusion and fission can both release energy

Light nuclei can release energy by fusing into more tightly bound products. Very heavy nuclei can release energy by splitting into intermediate-mass products. The energy release is

$$Q = (m_{\text{initial}} - m_{\text{final}})c^2. \tag{2}$$

Positive Q is exothermic. The binding-energy curve gives the trend; actual masses give the answer. Hydrogen-to-helium fusion converts about 0.7% of the initial rest mass into energy, far more per kilogram than chemical burning.

4. Radioactivity is a probability law

For a large sample, the decay rate is proportional to the number remaining:

$$\frac{dN}{dt} = -\lambda N, \qquad N = N_0 e^{-\lambda t} = N_0\, 2^{-t/T_{1/2}}. \tag{3}$$

Activity is $A = \lambda N$, measured in becquerels (decays/s). The half-life is $T_{1/2} = \ln 2 / \lambda$; the mean lifetime is $\tau = 1/\lambda$. The exponential is optional enrichment: the half-life form is enough for most Olympiad calculations.

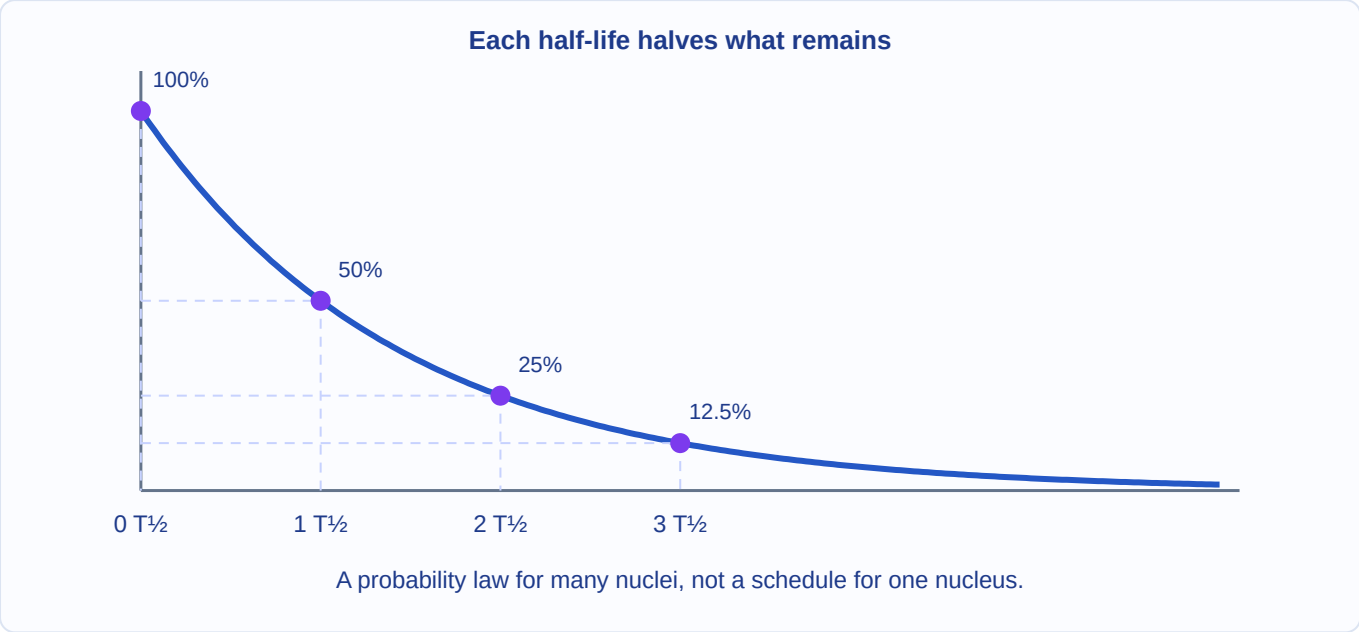


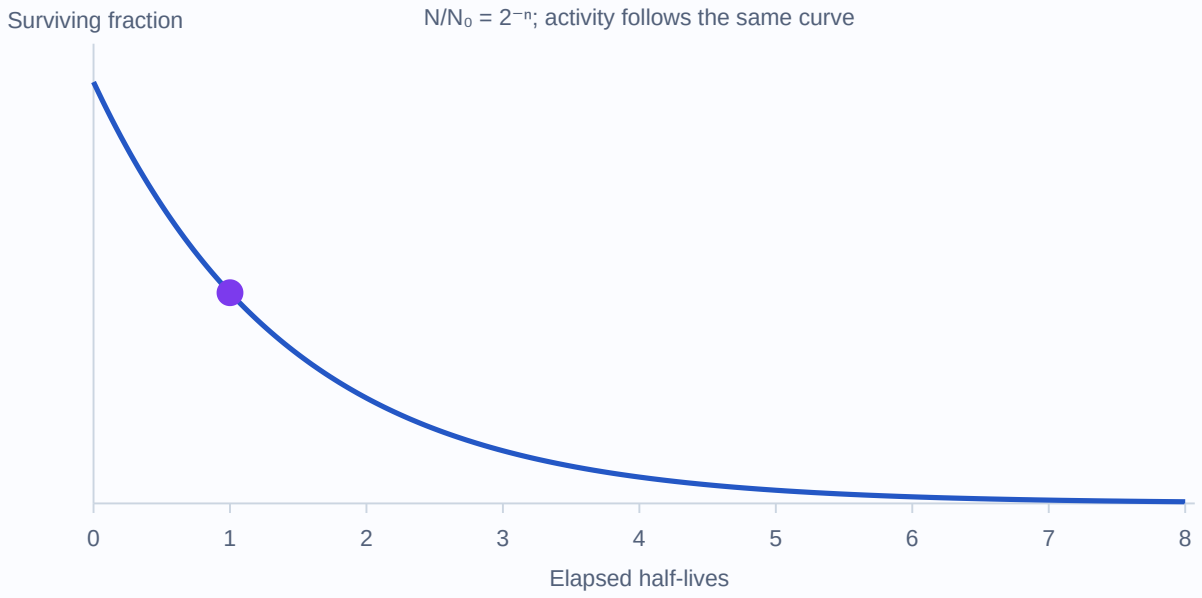
Figure 10. Activity is proportional to the number remaining, so it follows the same exponential. Individual decay times remain unpredictable.

INTERACTIVE EXPERIMENT

A clock made from a probability law.

Start with 10^{12} radioactive nuclei. The surviving number and activity decay together. The isotope changes the real-time scale, not the shape when time is measured in half-lives.

Nuclei remaining **5.000 × 10^11**	Surviving fraction **50.000%**	Activity **1.9166 Bq**
Elapsed time **5730.0 yr**	Decay constant **3.833 × 10⁻¹² s⁻¹**	Mean lifetime **1.443 half-lives**



A half-life is not a mean lifetime: τ = T½/ln 2 ≈ 1.443 T½. The activity is calculated in SI units from A = λN.

Try this. After three half-lives, one eighth remains—not zero. Switch isotope at the same number of half-lives and compare the activities.

Snapshot of the experiment. Open the HTML edition to change the controls.

5. Decay chains and equilibrium

A radioactive daughter can be produced while it is also decaying. Its abundance depends on both rates. If the parent is much longer-lived than the daughter, secular equilibrium can develop: daughter activity becomes approximately equal to parent activity even though the daughter contains far fewer nuclei.

For a parent p feeding a daughter d from an initially pure parent sample,

$$N_d(t) = \frac{\lambda_p N_{p0}}{\lambda_d - \lambda_p}\left(e^{-\lambda_p t} - e^{-\lambda_d t}\right). \tag{4}$$

This formula assumes a simple two-member chain and no initial daughter. It is enrichment beyond the minimum syllabus requirement.

6. Neutrinos: the missing energy carriers

Neutrinos interact weakly and can escape dense stellar cores. In beta decay they carry energy, momentum and lepton number. In the solar proton–proton chain, two neutrinos are produced per helium nucleus assembled. The photon luminosity therefore does not include all the nuclear energy release.

Neutrinos are marked qualitative in the IOAA syllabus, but simple counting and flux estimates are useful. A neutrino flux at distance d from an isotropic source is its production rate divided by $4\pi d^2$.

7. Dating and fuel clocks

Radiometric dating compares remaining parent with accumulated daughter, assuming a closed system and known initial daughter content. A stellar fuel clock divides available nuclear energy by luminosity. Neither clock is just a half-life: the relevant reservoir and loss rate must be identified.

Solar fuel estimate. If a fraction f of a star's mass can burn and a fraction ϵ of that mass becomes energy, $t \sim f\epsilon Mc^2/L$. For the Sun, $f \sim 0.1$ and $\epsilon \sim 0.007$ give about 10^{10} years at the present luminosity.

8. Check conservation and the energy destination

Balance A and Z before computing masses. Keep track of whether a quoted Q includes neutrino energy, annihilation energy or kinetic energy of a particular product. Activity is not the number of nuclei; it is the number remaining multiplied by a rate.

Common traps. After three half-lives, one eighth remains, not zero. Atomic masses include electrons. A reaction can be energetically allowed yet slow because of a barrier or a weak interaction. "Qualitative" neutrino coverage does not mean neutrinos are optional in the underlying physics.

9. Lecture + Problems: predict, calculate, check

Fifteen original problems: **[1]** foundation, **[2]** multi-step Olympiad reasoning, **[3]** challenge. Use the constants supplied in the lecture unless a question overrides them. Explain assumptions and keep units. Every problem has a worked solution.

[1] Problem 1 — Read and transform a nuclide. For $^{238}_{92}\mathrm{U}$, find Z and N. Write the daughter after alpha decay and after beta-minus decay of the original nucleus. What does gamma emission change?

Worked solution →

[2] Problem 2 — Weigh the binding. Using neutral atomic masses $m(^1H) = 1.00782503223u$, $m_n = 1.00866491595u$ and $m(^4He) = 4.00260325413u$, find helium-4's mass defect, binding energy and binding energy per nucleon.

Worked solution →

[2] Problem 3 — Four hydrogens to helium. Using the masses in Problem 2, find the total energy release for $4\,^1H \rightarrow\ ^4He$ with the required leptons included so atomic electrons balance. Find the mass-to-energy efficiency. Why is the photon/thermal energy slightly smaller?

Worked solution →

[2] Problem 4 — A fission fuel bill. Assume 200 MeV per fission of ^{235}U. Find the energy per kilogram and the fuel mass fissioned per day in a 1 GW thermal reactor. Use 235 g/mol.

Worked solution →

[1] Problem 5 — Three half-lives. An isotope has half-life 8 days and initial activity 80 MBq. Find activity and surviving fraction after 24 days.

Worked solution →

[2] Problem 6 — A radiocarbon age. A closed sample retains 72% of its original carbon-14. With half-life 5730 years, find its age.

Worked solution →

[2] Problem 7 — From atoms to becquerels. A sample contains 10^{15} cobalt-60 nuclei, half-life 5.27 years. Find its initial activity and its activity after 10.54 years.

Worked solution →

[3] Problem 8 — A daughter grows and decays. A parent with half-life 10 h feeds a daughter with half-life 20 h. Initially only parent is present. At what time is the daughter number largest? Find that number relative to the initial parent number and compare the two activities then.

Worked solution →

[3] Problem 9 — A reaction threshold. For $^{14}N + \alpha \rightarrow {}^{17}O + p$, use atomic masses 14.00307400443, 4.00260325413, 16.99913175650 and 1.00782503223 u. Find Q. Estimate the alpha kinetic-energy threshold for a stationary nitrogen target using $E_{\text{thr}} \simeq -Q(1 + m_\alpha/M_N)$.

Worked solution →

[3] Problem 10 — Count solar neutrinos. The Sun's photon luminosity is $L_\odot$. Suppose 26.2 MeV per helium nucleus appears as photon/thermal energy and two neutrinos are produced. Estimate the helium production rate and neutrino flux at Earth, ignoring oscillations.

Worked solution →

[3] Problem 11 — Fusion energy sharing. For $D + T \rightarrow {}^4He + n$, use masses 2.01410177812, 3.01604928199, 4.00260325413 and 1.00866491595 u. Find Q. If the initial kinetic energy is negligible, estimate the neutron and alpha kinetic energies.

Worked solution →

[1] Problem 12 — Why cooling continues. Why does a fission reactor keep producing heat after the chain reaction is stopped? What does this imply for cooling systems?

Worked solution →

[2] Problem 13 — Parent and daughter. A stable daughter is produced by a parent with half-life 12 years. A closed rock now has parent:daughter ratio 1:3 and had no initial daughter. Find its age. How would initial daughter affect the inference?

Worked solution →

[3] Problem 14 — Secular equilibrium. A parent of half-life 10^9 years feeds a daughter of half-life 10 years. After equilibrium is established, compare their activities and numbers. Why can a tiny daughter abundance have the same activity?

Worked solution →

[3] Problem 15 — The Sun's nuclear clock. Assume 10% of the Sun's mass can fuse hydrogen to helium with efficiency 0.007. Estimate the main-sequence fuel timescale at the present luminosity. Compare with the gravitational contraction timescale.

Worked solution →

Further practice and provenance

The 15 exercises above and their solutions are **original MAIS PAC problems**, inspired by the skills in this syllabus topic. They are not copied from the papers below. The earlier club lecture compiled these official past-paper identifiers for further practice; only identifiers are reproduced, not copyrighted question statements or official solutions. Round and question labels retain the earlier compilation's notation and should be checked against the official paper.

- IOAA 2008 · TH · S4
- IOAA 2009 · TH · P10
- IOAA 2009 · TH · P11
- IOAA 2010 · TH · S5
- IOAA 2011 · TH · S14
- IOAA 2012 · TH · S6
- IOAA 2013 · TH · S3
- IOAA 2014 · TH · P3
- IOAA 2014 · TH · P6
- IOAA 2015 · TH · S12
- IOAA 2017 · TH · T6
- IOAA 2018 · TH · T11
- IOAA 2023 · TH · Q8
- IOAA 2024 · TH · T4
- IOAA 2025 · TH · T07
- IOAA 2025 · TH · T10

Sources and scope. Topic coverage follows the IOAA syllabus. Official problems and constants are available through the IOAA past-paper archive. The derivations, diagrams, experiments and numerical teaching examples in this handout were prepared independently for MAIS PAC. Calculator-level methods are emphasized; optional calculus connections are explicitly identified.